

Lab 2: Proportional Control

Overview

The Mayan calendar may have been wrong about the world ending for the rest of us back in 2012, but tragically, for a cat in the Netherlands named Orville, the apocalypse did arrive that year in the form of an oncoming car. Orville was survived by his brother, Wilbur, and human roommate, Bart Jansen, who decided to honour Orville's memory by turning him into a pioneer of flight, like the Wright brother who gave him his name. With the help of an engineer, Arjen Beltman, and a taxidermist, Orville became the Orvillecopter and was finally able to take to the skies among the birds that so fascinated him in life.

Now, you get to have a (virtual) kitticopter to call your own! In this lab, you will design a proportional controller to help your flying feline track a position input according to a set of specifications.



This assignment will take place over **two** sessions in the lab: first, you will perform tests to determine a model of your cat, then you will test your controller on the system. **Both sessions have required prep work that you MUST do beforehand. If you haven't done your prep work, you cannot attend the lab.**

Session 1: Getting to Know Your Cat

All cats are beautiful, but some cats are more aerodynamic than others. Your first objective is to determine a suitable model for your kitticopter that you can use to design your controller. The parameters of your cat are unique to you. Unlike the first lab, you will be given a limited number of chances to confirm the accuracy of your transfer function of your cat, however, you are still advised to validate your model through simulations. You will also be using the same system in the next lab, so ensure that you model it as accurately as you can to save yourself two labs' worth of trouble.

Preparation

Consider the following information about your kitticopter:

- The kitticopter is acted upon by three forces: rotor thrust $f_r(t)$ (the input to the system), aerodynamic drag and gravity.
- You may assume that all parameters are normalised to the mass of the kitticopter (i.e. unit mass).
- You may model aerodynamic drag as being directly proportional to the velocity of the kitticopter, but in the opposite direction. You can use b to stand in for the proportionality constant.
- You may model gravity as a constant disturbance at the **input** to the system. Let $g = 9.81 \text{ m/s}^2$.
- The kitticopter is equipped with a sensor that perfectly measures its **position** $y(t)$. It may be easier to work in terms of velocity to identify your system parameters. You can do this by differentiating the output of your system.
- To allow for electronic control, the kitticopter's position must be converted from a value in meters to a voltage by the feedback sensor. The relationship between the position and the voltage is a simple scalar k_s that needs to be determined experimentally. The scalar is unique to each student.

- The command input from the controller to the kitticopter will also be in the form of a voltage. This voltage controls the angle of attack of the rotor blades, $\theta(t)$, which determines the thrust. Your controller can output voltages between 0 and 5 V, which is calibrated such that the middle of the voltage range corresponds to the thrust that perfectly negates gravity.

Based on the given information, please complete the following tasks **before** attending session 1 of your lab.

- Draw a block diagram of the kitticopter system, representing all of the given information.
- Derive a differential equation describing the acceleration of the kitticopter.
- Derive the open-loop transfer function $G(s)$ relating the **velocity and rotor thrust** of the kitticopter, and the closed-loop transfer function $G_{cl}(s)$ relating the **position** to the **given setpoint $x(t)$** for the overall system.
- Derive closed-loop transfer functions describing (i) the position error in relation to the setpoint, and (ii) the effect of gravity on the position output. Determine the **type number** of the system and describe the tracking performance you expect to be able to achieve using proportional control with respect to position setpoints (i.e. will the error and effect of the disturbance be infinite, finite or zero?)
- OPTIONAL: Create a Simulink model of the kitticopter system and bring it with you to the lab. This is not essential, but it will allow you to simulate your proposed model during the session rather than afterwards.

Briefly write up these tasks and present them to Mr. de Maar or a tutor at the start of the lab session. It would be a good idea to check your work with a tutor well before your session, so you can be sure that you are modelling the system correctly. Each session is limited to **an hour**, so you don't want to waste valuable lab time redoing your prep work if you did not do it correctly. Ensure that you bring your **breadboard** (as well as some connecting wires and tools with you), as you will be using it to construct the input circuit for this experiment.

Instructions

The simulated kitticopter system operates in a similar way to the spring-mass-damper system you used in the previous lab, but this time, you will be giving it inputs that you create using a physical voltage rather than ideal simulated inputs. These will be fed into the computer through an analogue-to-digital converter (ADC). The simulated kitticopter responds to values between 0 and 5 V.

You will build a simple potentiometer circuit using your breadboard to control the voltage given to the ADC. Instructions for how to do this are provided in the attached lab manual. The potentiometers will be provided to you in the lab.

The goal of this session is to complete the transfer function model $G(s)$ by determining the unknown aerodynamic constant b of your kitticopter by performing a step test.

Hints:

- It may be easier to work at the velocity level for these open-loop tests, as the relationship between the velocity output and rotor thrust input should be first-order. You can get back to the velocity output by taking the derivative of the position response you obtain.
- You want to start your step from initial rest, but because of gravity, this does not necessarily mean zero input voltage. Zero input voltage will, in fact, result in your poor cat plummeting to earth (luckily, they always land on their feet, so no harm done). You need to find the input voltage where gravity is perfectly counteracted by the rotor thrust, so the cat hovers at a steady height. It is

important to note this value, as it is effectively the “zero” point for your controller, and you will have to offset the controller output by this value to achieve the expected results.

- Because of the hand-made input step and imprecise process of taking the derivative of the position digitally, you are advised to repeat the step test several times. The average of the resulting system parameters will probably be closer to the truth than the results of any individual test.

Clearly document the tests you perform as part of your lab report.

Session 2: Proportional Controller

Now that you have a good model of your system, you are ready to do some actual control!

Specifications

Your proportional controller should aim to achieve the following performance specifications:

- 1) Tracking of position inputs with $>92\%$ accuracy (i.e. the tracking error and effects of disturbances must be $<8\%$).
- 2) Overshoot of less than 30%.
- 3) Robustness to uncertainty of up to 10% in the aerodynamic constant of the system. (Since you are determining this parameter experimentally, you should make sure your design works even if you don't get it precisely “correct”.)
- 4) Robustness to a tolerance of 10% in the components used to assemble the controller.

You may also improve on these specifications if you are able to. The inputs your controller can deliver are limited to 10V by the power supply, so this will be the ultimate restriction on what you can achieve.

Preparation

Completion of the following activities is essential before you may attend session 2:

- 1) **Session 1:** Present your write-up of the first session to the tutors or Mr De Maar, showing that you have obtained a complete model of the system.
- 2) **Transfer Function check:** Once you have derived your transfer function, you may check the accuracy of your model as laid out in Appendix A2 of this document. This is to ensure that your model is not so incorrect that you are unable to design a proportional controller capable of controlling your system and meeting the design specifications. NB: You will only be given **3 attempts** at doing this.
- 3) **Proportional controller design:** You can use any methods (in the time and/or frequency domain) learned in class to design your controller. Clearly explain your reasoning and show any graphics or calculations needed to support your decisions. You may also wish to perform preliminary simulated tests of your controller's performance using MATLAB or other software. Present your design at the start of the session.
- 4) **Controller build:** Construct your controller **on Veroboard** and present it at the start of the session. The Veroboard and op-amps will be provided to you by Mr De Maar. You will have to get any other components you may need (such as resistors and capacitors) from the White Lab. You may bring a controller on the breadboard, but your mark for the controller demonstration portion of the lab will then be **capped at 50%**.

Hint: Appendix A1 of this document includes instructions for how to build a proportional controller using an op-amp, and a helpful recap of how op-amps work.

Instructions

The aim of this session is to demonstrate that your controller meets all the given specifications. You may perform any suitable tests to do so. You will find instructions for how to connect your controller to the simulated system in Appendix A1 of this document.

Demo: When you are ready, Mr De Maar will also take you through a standard step testing process for you to demonstrate your controller. Please include a screenshot of your controller tracking the setpoint in the appendix of your report. He will **take your controller** at the end of the demo, so please ensure you have completed everything you need it for before demonstrating.

EEE3094S Lab 2 Report

Your report has a maximum page limit of **15 pages** (not including Appendices). This is not a target, and having more pages does not mean your marker will be more impressed with you. A 2–3 page report is perfect for what we've asked you to do.

Please follow the report-writing guidelines given in Lab 1. As before, the point of this reporting exercise is to prepare you for future projects in your degree and your life as a professional engineer, so we expect a high standard of professional presentation. This includes but is not limited to:

- NO photos of handwritten work
- **NO Excel plots.** Use MATLAB, Python or really any other software.
- Plots and images that are not obviously screenshotted, pixelated, or badly scaled.
- Figures must have captions, annotations and be referred to in the text. They need to serve the text of the report and be integrated into it effectively.
- Clear but concise descriptions of your process and calculations. A reader should be able to understand and replicate your work from your descriptions.
- It is not essential to use LaTeX, but you will be more likely to come out with a neat and professional report if you do.

A LaTeX template has been provided to you.

Your report should include the following sections:

System Modelling

This section explains how you arrived at the format of the model you are using for your system (i.e. what is covered in the pre-lab). This is where you determine the “known unknowns” of your system: the parameters you need to find values of experimentally during system identification.

Some things we would expect to see in this section include:

- A free-body diagram illustrating the forces acting on the kitticopter
- A block diagram of the whole system showing the various signals
- A differential equation describing the motion of the kitticopter
- A transfer function derived from the differential equation

You should explain your modelling decisions using information that is given to you about the system.

System Identification

This section should include:

- Plots of your step test(s) and results.
- Well-described calculations of your system parameters based on your step tests.
- Validation of your model

Controller Design

This section should include:

- A description of your controller design process, supported by calculations.
- Tests of your controller in simulation.
- Details of the physical design of your controller, e.g. a schematic, and photos of your final board.

Controller Tests

This section should include:

- A description and plots of the tests you performed to evaluate your controller experimentally
- An evaluation of your controller's performance against the given specifications, culminating in an ultimate conclusion about whether your controller worked as intended.
- Recommendations of how to improve the performance of your controller (if applicable).

Rubrics

We will apply a 20% per day late penalty.

Category	Fail [0-10]	Passable [10-12]	Good [12-15]	Great [15-20]
System Identification [20]	No/ an incorrect attempt to explain conversion to velocity. Step test was not done or conducted in a way that was flawed to the point of being useless. OR Results are not displayed, or displayed in a way that is difficult to interpret. Results are not validated.	No/ a weak or incorrect explanation of the conversion to velocity OR student attempts to work with second-order system, but displays some knowledge of system identification. Step test was done mostly correctly, but there may be some errors (e.g. cutting the response off too early to obtain a true final value, using a bad input step) OR there are major problems with how the results are displayed (e.g. unlabeled graphs, marker or colour choices that are hard to read, ugly screenshots.) Results are not validated/ validation is weak and/or poorly documented.	Process of converting to velocity to give a firstorder response was explained but there are some minor flaws in the explanation or documentation. Step test was done correctly, but there may be some minor errors (e.g. not repeating the test multiple times, small miscalculations in the analysis) OR minor problems with how the results are displayed (i.e. plots that are readable but not as clear as they should be due to poor scaling/ tick choice/ cosmetics/ etc.) An attempt to validate the results was made but there minor issues with how it is executed or documented.	Process of converting to velocity to give a firstorder response was explained well. Step test was done correctly and repeated multiple times to account for uncertainty. Results were then aggregated logically to obtain the parameters. Model was validated sensibly and the experiments, results and conclusions were well documented (with well annotated figures).

Controller Design [20]	Analysis/ the design process is totally incorrect, not based on anything learned in class or absent/ impossible to follow in the report. There is little to no engagement with the specifications.	The analysis/ design process applies some concepts learned in class but they are not wellunderstood, and there are fundamental mistakes in the approach. The design is linked to the specifications (i.e. they are converted into pole placements/ gain values) but it is either incorrect OR there is no/ a weak attempt to assess the plausibility of the specifications.	The analysis/ design process is logical and wellgrounded in the theory of the course. There may be some minor errors e.g. calculation mistakes, or small misunderstandings. Analysis is supported by some evidence. (Figures may have some minor flaws but are still readable.) The design is strongly linked to the specifications and the plausibility of the specifications is considered. There may be some small problems with this analysis or how it is documented. The design is supported by simulations. There may be	The analysis/ design process is logical and wellgrounded in the theory of the course, demonstrating a strong understanding, and calculations are accurately done. Analysis is supported by strong evidence that is welldocumented. If MATLAB or similar software is used, the outputs are still validated with manual calculations and correct explanations. The design is strongly linked to the specifications, and the plausibility of the specifications is considered. This analysis is clearly explained and well-
	The design is not supported by simulations/ the simulations are performed poorly. MATLAB or similar software is used incorrectly, OR in a way that appears somewhat correct, but the outputs are not validated or are weakly validated. No/ very little consideration is given to the robustness of the design.	The design is not supported by simulations/ the simulations are performed poorly. MATLAB or similar software is used in a way that appears mostly correct, but the outputs are not validated or are weakly validated. No/ very little consideration is given to the robustness of the design.	some small problems with how these are executed or documented. If MATLAB or similar software is used, the outputs are still validated with manual calculations and correct explanations. Limited consideration is given to the robustness of the design.	documented. Reasonable new specifications are proposed. The design is supported by simulations. These are executed sensibly and documented effectively. Consideration is given to the robustness of the design.
Controller Test (Demo) [20]	See controller demo rubric below			

Controller Evaluation [20]	Results are not analyzed/ analysis is largely incorrect or incoherent. There are no figures or the figures are unintelligible/ not referred to in text. The performance of the controller is not evaluated or the evaluation is mostly incorrect. The original specifications are not referred to. No recommendations were made where they should have been, OR the recommendations made were not based on any evaluation of the controller's performance.	Analysis of the results goes in the right direction but is weak or with major flaws in understanding OR is not supported by figures/ figures are not clear. Conclusions about the performance of the controller are basically correct but incomplete or not well-grounded in the results. There is no attempt or a weak attempt to link the evaluation to the specifications. There was an attempt to make recommendations based on the conclusions, but they are impractical, OR do not address the specifications and limitations of the control design.	Results were analyzed in a way that displays some understanding of the course material and a link to the results, but there may be some minor flaws in the interpretation OR in the way that the data is presented OR the figures are present and readable but not tied to the analysis. Conclusions about the performance of the controller are mostly logical and well-supported by the results, and link back to the original specifications, but there may be some minor errors or things left out. Practical recommendations were made based on the conclusions drawn above, where applicable, but there may be some minor errors or things left out, OR they partially link back to the specifications and/or do not fully address the limitations in the control design.	Results were analyzed logically, showing understanding of the course material and analysis was linked to clear figures. Conclusions about the performance of the controller are logical and well-supported by the results in the paper. The evaluation clearly links the results back to the specifications of the project, and the limitations of the controller design. Recommendations were made based on the conclusions drawn above, where applicable and are well-supported by the results of the paper. The recommendations are practical and seek to improve the controller's performance based on the specifications of the project and/or address the limitations present in the control design.
Report [20]	Report is unintelligible, very unprofessional or has a lot of missing information.	Report has all the basic information but there are some spelling, grammar or formatting issues.	Report presents all the basic information clearly and professionally. The key steps of the method are described, but some details may have been left out.	Report is clear and professional. Methods are described in sufficient detail to be fully repeatable. Decisions are well-motivated, with clear links drawn between text and figures.

EEE3094S Lab 2 Controller Demonstration 2025

Student Name		Student Number	
---------------------	--	-----------------------	--

Controller Demonstration Marking Guidelines (Total Marks 20)						
Marks	0	1	2	3	5	MARK
Set Point Tracking	No Tracking or Large Error	Within 1m of final value	Within 0.8m of final value	Within 0.6m of final value	Within 0.4m of final value	
Overshoot	Unlimited / Very Large		Within 50%	Within 40%	Within 30%	
Simulated Plot for an input Step of 5m	No Plot		Plot shown not within the specifications		Plot shown within the specifications	
Circuit diagram	No circuit diagram	Partial circuit diagram	Circuit diagram not fully labelled		Fully labelled circuit diagram	
Penalty Marks	Controller Output: High amplitude oscillations of $> 0.4V$ 2 Mark penalty	On/Off Bang-Bang Controller Type 8 Mark penalty	Using more than 4-Op Amps 3 Mark penalty	Other: Mark capped at 50% if implemented on breadboard		
TOTAL MARKS						

Assessor Name:

Signature:

Date: